

# Vol 9 Chapter 8 — BaTiO<sub>3</sub> 137-Plane: BST's Signature Condensed-Matter Falsifier

Keeper (author pass — deep math/physics revision)

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## Chapter 8 — BaTiO<sub>3</sub> 137-Plane: BST's Signature Condensed-Matter Falsifier

*Extended-reach chapter (May 2026), identification tier. Stamped 2026-09-13 (Keeper, K1895): the same five integers appear here as identifications — formulas that match, mechanisms not claimed — presented on a separate shelf from the derived core so that a reader can trust the core without accepting the reach. Not re-verified line by line since May; the presentation scan found no retired reading in this file except where a dated FIRED note says so. Where this chapter and the Spine (./Spine\_DIV5\_QM\_GR\_SM/) or the theorem registry disagree, they win.*

### Level 1 — one sentence

This chapter is load-bearing: barium titanate (BaTiO<sub>3</sub>) in its 137-plane crystallographic configuration is predicted by BST to exhibit a substrate eigentone — a resonance at a specific frequency derivable from the BST primary  $N_{\max} = 137$  — and the experiment to test this (~\$25K, standard solid-state laboratory, ~6-12 month timeline) is BST's lowest-cost lab-accessible falsifier.

### Level 2 — graduate-physicist precision

#### 8.1 BaTiO<sub>3</sub> basic properties

Barium titanate BaTiO<sub>3</sub>: ABO<sub>3</sub> perovskite structure. Discovered in 1940s, widely used as ferroelectric, piezoelectric, and capacitor dielectric.

Phase transitions (with decreasing temperature): - Cubic Pm $\bar{3}$ m (above ~393 K) — paraelectric - Tetragonal P4mm (~278 to ~393 K) — ferroelectric (room temperature phase) - Orthorhombic Amm2 (~183 to ~278 K) — ferroelectric - Rhombohedral R3m (below ~183 K) — ferroelectric

Spontaneous polarization (tetragonal phase): ~26  $\mu\text{C}/\text{cm}^2$ . Used in non-volatile memory (FeRAM).

Dielectric constant: extremely high near phase transitions (~10,000 at Curie point).  
Used in compact capacitors.

## 8.2 Why “137-plane”

BaTiO<sub>3</sub> crystallographic planes are conventionally labeled by Miller indices (*hkl*). The “137-plane” refers to the (137) plane in the cubic phase — a high-Miller-index plane that engages the substrate’s natural integer  $N_{\text{max}} = 137 = N_c^3 \cdot n_c + \text{rank}$ .

BST substrate-mechanism reading: the substrate “prefers” configurations whose geometric structure matches BST primary integers. The (137) plane of BaTiO<sub>3</sub> provides a specific lattice configuration where the substrate’s  $N_{\text{max}} = 137$  K-type structure engages directly, producing a resonance condition (eigentone) at a derivable frequency.

## 8.3 The substrate eigentone prediction

Prediction: BaTiO<sub>3</sub> in (137)-plane orientation exhibits a substrate-natural resonance at frequency  $f_{137}$  derivable from BST primaries. The exact  $f_{137}$  value depends on substrate operator zoo calibration (SP-31-1 and SP-31-6 program); current estimates are in the GHz-THz range, but the precise value requires the substrate Hamiltonian closure (K52a Sessions 6-14).

Observable: enhanced microwave or far-IR absorption / emission at  $f_{137}$  in (137)-oriented BaTiO<sub>3</sub> samples vs other orientations as control.

Q-factor: substrate eigentones predicted to have high Q (~10<sup>3</sup>-10<sup>4</sup>) because substrate resonance is sharp.

## 8.4 Experimental design

Apparatus: - (137)-oriented BaTiO<sub>3</sub> single crystal (commercial; ~\$5K per crystal) - Microwave/THz spectrometer (existing solid-state lab equipment) - Control: (100)-oriented crystal of same composition

Protocol: - Measure dielectric/absorption spectrum 1 GHz - 5 THz on both orientations - Look for excess feature in (137) sample not in control - Specific predicted frequency from BST primaries

Budget estimate: ~\$25K total (crystals + spectrometer time + analysis). Achievable in 6-12 months.

Sites: any solid-state physics lab with microwave/THz capability. NIST, NPL, PTB national-standards labs; many university labs.

## 8.5 Outcomes

Success (eigentone observed at predicted  $f_{137}$  in (137), absent in control): - BST substrate framework gets a precise empirical confirmation - Substrate engineering

becomes practical experimental discipline - Other substrate eigentone predictions (photonic crystal Ch 9, atomic-clock decoherence) become higher-priority

Failure (no eigentone observed at predicted  $f_{137}$ ): - BST substrate framework refuted for condensed-matter sector - Substrate-mechanism predictions for materials need fundamental revision - BST other-sector predictions (Bell sub-Tsirelson, particle physics) survive but lose substrate-mechanism justification [FIRED — bracket 10:09 2026-09-13 (Grace, R143 G2; register E4, certified K1893): the sub-Tsirelson ceiling / SCMP Layer-1 prediction quoted here was refuted by Poh et al., PRL 115, 180408 (2015):  $S = 2.82759 \pm 0.00051$ ,  $41.9\sigma$  above BST's 2.806. This sentence is a May-2026 record, not a standing claim; nothing else on the line is changed.]

This is a clean go/no-go test. Decisive.

## 8.6 Why this matters

Most physics frameworks have either: - Beautiful math but no lab-accessible falsifier (string theory) - Working empirical fit but no underlying theory (Standard Model) - Specific predictions but too expensive to test (GUT proton decay)

The BaTiO<sub>3</sub> 137-plane test is the rare case of a: - New substrate-mechanism framework - With specific BST-derived prediction (not curve-fit) - Lab-accessible at modest cost (\$25K) - Decisive go/no-go outcome

If BST is wrong, this test exposes it quickly. If BST is right, this test launches the substrate engineering program seriously.

## 8.7 Status

SP-30-2 Boundary condition design (BST task #196, pending): formal experimental specification document for outreach to experimental groups. Outreach targets to be identified by Casey send-signal.

## 8.8 K-audit anchors

- Vol 0 Ch 5:  $N_{\max} = 137$  derivation
- SP-30 (substrate engineering program): parent task #185
- SP-30-2 (task #196): BaTiO<sub>3</sub> experimental design
- Operator zoo SP-31-1: substrate Hamiltonian (needed for precise  $f_{137}$ )

## Level 3 — 5th-grader accessibility

Killer test for BST: Take barium titanate (BaTiO<sub>3</sub>, a common ferroelectric used in capacitors and memory chips). Orient the crystal so its (137) plane is the active surface. BST predicts this configuration will ring at a specific frequency — a substrate eigentone. Measure the response with standard microwave/THz spectroscopy (\$25K total cost, 6-12 months).

- If the eigentone is there at the predicted frequency: BST gets a sharp empirical confirmation; substrate engineering becomes real.
- If no eigentone is there: BST's substrate-mechanism for condensed matter is wrong.

The number 137 isn't random — it's the BST primary  $N_{\max} = N_c^3 \cdot n_c + \text{rank} = 27 \cdot 5 + 2$ . Any solid-state physics lab can do this experiment.

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### What comes next

Chapter 9 develops the photonic crystal ~\$10K eigentone test — even cheaper.

### Where to look this up

- BaTiO<sub>3</sub> phase diagrams: Lines and Glass, *Principles and Applications of Ferroelectrics*
- BST: Vol 0 Ch 5 ( $N_{\max}$ ); SP-30-2 task #196